Term 3 Robotics Challenge

Trial Runs #1 Checklist

1. Mechanical Design [10 pts]

In this test, your team must demonstrate the following:

1. Your robot satisfies the **designated dimensions** **[1 pt]**
2. Your robot's **suspension and drivetrain** can clear all the relevant obstacles (hold on the ramp without sliding down, drive over the planting holes without getting stuck) by pushing your robot over the different parts of arena manually **[2 pts]**
3. Your **'planter'** ('hopper') **mechanism** in operation **[2 pt]**:

- You must be able to load all 5 'seeds' at once and demonstrate that they can be deposited reliably and repeatedly (*without getting stuck, mechanism getting jammed etc.*) by activating the planter's actuator/mechanism manually (*e.g. by rotating some part of mechanism with your fingers or providing voltage to the motor*)
- Please note that you do **not** need to read any RFID tags here or verify that the seed has been planted correctly to pass this test. This is simply testing the mechanism.

4. **Reviving** ('tapping') **interface** in action **[2 pt]**:
    - Normally, an LED above your robot should be red.
    - The LED above your robot turns **green** when the interface button is pressed;
    - It goes back to red once the button is depressed.
    - You can use either two separate LEDs or a single RGB LED.
5. You have a **battery compartment** where you can place it securely (*so that it does not fall out if your robot is flipped upside down*) **[1 pt]**
6. Maturity of your robot's prototype: It should be mechanically sound, no gaffer tape, hot glue etc. **[2 pt]**

2. Electronics Design [10 pts]

In this test, your team must demonstrate the following:

1. Mechanical **kill switch** (button) in operation [1 pt]:
    a. Robot starts (e.g. moves) when the button is pressed;
    b. Robot stops when the button is pressed again.
    c. The LED on top of the robot (tapping mechanism) is **blinking red** when stopped.
2. **Wi-Fi kill switch** in operation: Robot stops when a ‘Stop’ sequence is communicated via UDP [2 pts]**:**
    - The LED on top of the robot (tapping mechanism) must be blinking red when stopped.
3. Your ability to control **robot’s speed and heading** [2 pts]**:**
    - Make your robot go forward at different speeds
    - Make your robot **turn left and right** by 90 degrees
    - Make a **U-turn**.
4. **Reflectance sensors** can detect the strip (for line following):
    - Display sensor readings via Arduino IDE or any other development environment you are using (live demonstration) [2 pts]
5. **Distance sensors** can successfully detect the wall:
    - Display sensor reading (distance) via IDE (live demo) [1 pt]
6. **RFID** sensor in action: You can detect and display the tag’s ID [1 pt]
7. General electronics **sub-system maturity** [1 pt]:
    - Your robot is using a soldered strip (perforated) board instead of a breadboard
    - Sustainable cable management: No ‘free floating’ components suspended by wires, the use of cable ties, wires do not get stuck under/within the mechanism(s) and moving parts, etc.